14 Bipolar Junction Transistor (BJT)

14.1 Motivation: From Two-Terminal to Three-Terminal Amplification

In our study of the PN junction, we discovered something remarkable. When we forward-bias the junction, the minority carrier concentration at the edge of the depletion region becomes:

$$\delta p(x_n) = p_{n0}\left(e^{qV/kT} - 1\right)$$

This is an **exponential function** of the applied voltage. A small change in voltage produces a huge change in carrier concentration—and therefore a huge change in current. This is, in essence, an **amplification mechanism**: a small input signal (voltage) controls a much larger output signal (current).

The Problem: No Third Terminal

If a single PN junction can amplify, why do we need a more complex device? The answer lies in **circuit practicality**:

- A two-terminal device (diode) has the same two terminals for both input and output
- You cannot independently control the input and monitor the output
- There is no **ground** or reference terminal to separate input from output
- The input signal and output signal are intrinsically mixed together

What we need is a **three-terminal device** where:

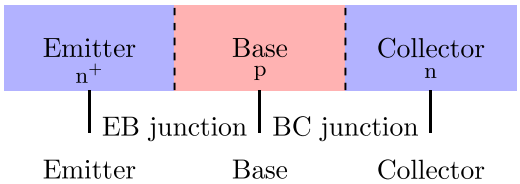
- One terminal serves as the input (signal applied here)
- Another terminal serves as the output (amplified signal taken from here)
- The third terminal serves as a common reference (ground)

This allows the input and output circuits to be separate while still allowing the input to control the output. This is exactly what the **Bipolar Junction Transistor (BJT)** provides—a **controlled PN junction**.

14.2 Device Structure: Three Layers, Two Junctions

The BJT consists of three alternating layers of doped semiconductor, forming two back-to-back PN junctions. There are two types:

- **npn transistor**: emitter and collector are n-type, base is p-type
- **pnp transistor**: emitter and collector are p-type, base is n-type



npn BJT Structure

14.3 The Four Modes of Operation

With two junctions (emitter-base and base-collector), each can be either forward-biased (FB) or reverse-biased (RB). This gives four distinct operating modes:

Mode	EB Junction	BC Junction	Application
Active (Forward)	Forward bias	Reverse bias	**Amplification**
Cutoff	Reverse bias	Reverse bias	Switch (OFF)
Saturation	Forward bias	Forward bias	Switch (ON)
Inverse (Reverse active)	Reverse bias	Forward bias	Rarely used

14.4 Understanding the Base Current: Recombination and Charge Neutrality

A critical question: **Where does the base current come from?**

In the active mode, the emitter injects electrons into the base (for npn). These electrons are minority carriers in the p-type base. As they diffuse across the base toward the collector, some of them recombine with holes (the majority carriers in the base).

Charge Neutrality Demands Base Current

When an electron recombines with a hole in the base, that hole is **annihilated**. To maintain charge neutrality in the base, the missing hole must be replenished. This is accomplished by the **base current** I_B:

- Electrons enter the base from the emitter
- Some recombine with holes, removing those holes from the base
- The external base current supplies holes to replace those lost to recombination
- Charge neutrality in the base is maintained at all times

The base current I_B is therefore **proportional to the recombination rate** in the base:

$$I_B = \frac{Q_b}{\tau_b} = \frac{\text{stored charge in base}}{\text{recombination lifetime}}$$

The larger the base current, the more recombination is occurring. Since I_B is typically much smaller than I_C, this means **most injected electrons reach the collector without recombining**.

14.5 Physical Mechanism of Amplification

The amplification mechanism proceeds as follows:

1. **Emitter injection:** The forward-biased EB junction injects electrons from emitter into base. The injected electron concentration at the edge of the base is:

$$\delta n_b(0) = n_{b0}\left(e^{qV_{BE}/kT} - 1\right) \approx n_{b0}e^{qV_{BE}/kT}$$

This is the same exponential relationship we saw in the PN junction.

2. **Diffusion across the base:** The base is extremely thin ($W_b \ll L_n$). Most injected electrons diffuse to the base-collector junction without recombining. This is governed by the diffusion equation we derived earlier:
$$\frac{\partial \delta n}{\partial t} = D_n \frac{\partial^2 \delta n}{\partial x^2} - \frac{\delta n}{\tau_n}$$

3. **Collection:** The reverse-biased BC junction creates a strong electric field that sweeps the diffusing electrons into the collector. The current reaching the collector is:
$$I_C \approx I_E \quad \text{(most electrons survive)}$$

4. **Base current:** The small fraction of electrons that recombine in the base must be replaced. This constitutes the base current:
$$I_B = I_E - I_C$$

5. **Current gain:** Because $I_C \approx I_E$ and I_B is small:
$$\boxed{\beta = \frac{I_C}{I_B} = \frac{I_C}{I_E - I_C} = \frac{\alpha}{1-\alpha}}$$
where $\alpha = I_C/I_E$ is the **common-base current gain** (typically $0.95 - 0.99$).

6. **Amplification in action:** A small change in base current produces a large change in collector current:
$$\boxed{i_c = \beta\, i_b}$$
This is the fundamental amplifier equation of the BJT.

The Role of Base Width

The transport factor α_T (fraction of injected electrons that reach the collector) depends on base width:
$$\alpha_T \approx 1 - \frac{W_b^2}{2L_n^2}$$
where $L_n = \sqrt{D_n \tau_n}$ is the diffusion length. This shows:

- **Thinner base** (W_b small) $\rightarrow$ less recombination $\rightarrow$ higher $\alpha \rightarrow$ higher β
- Typical base widths are $0.1 - 1\mu$m, much smaller than L_n (typically $10 - 100\mu$m)
- This is why the base must be extremely thin for good amplification

14.6 Understanding Saturation: The ON State of the Switch

What is $V_{CE,sat}$? $V_{CE,sat}$ is the **collector-emitter saturation voltage**—the minimum voltage across the transistor when it is fully turned on.

Physical explanation of saturation: When the base-collector junction becomes **forward-biased** (as well as the EB junction), the electrostatic conditions change fundamentally:

1. **Reverse bias of BC junction** in active mode creates a depletion region that sweeps carriers from base to collector. The electric field in this region is strong and directed to accelerate electrons from base into collector.

2. **Forward bias of BC junction** in saturation reduces this electric field to nearly zero. The depletion region shrinks, and in extreme cases, the built-in potential barrier at the BC junction is overcome.
3. **Consequence for carrier flow:** Without the strong collecting field, electrons that diffuse to the base-collector boundary are no longer efficiently swept into the collector. They accumulate at the junction, and some may even be injected backward from collector into base.
4. **Collector current becomes limited by external circuit:** The transistor can no longer pull more current regardless of base drive. The collector-emitter voltage drops to its minimum possible value:

$$V_{CE,sat} = V_{BE,sat} - V_{BC,sat} \approx 0.7 - 0.5 = 0.2 \text{ V (for silicon)}$$

The Electrostatics of Saturation

In saturation:

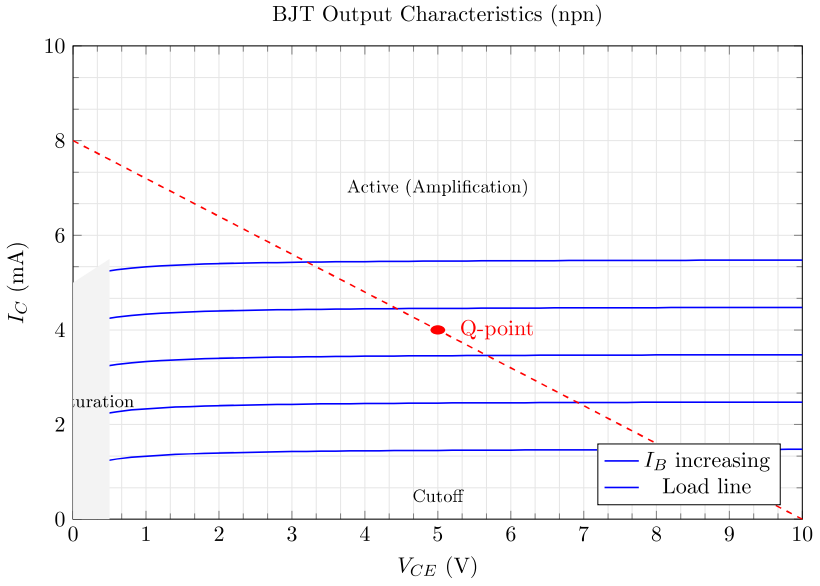
- Both junctions are forward-biased
- The base-collector depletion region vanishes
- Electrons can flow from emitter to collector, but also from collector to base (reverse injection)
- The collector current is determined by the external circuit (V_{CC} and R_C), not by I_B
- V_{CE} is the difference between two forward-biased diode voltages:

$$V_{CE} = V_{BE} - V_{BC}$$

  Since both V_{BE} and V_{BC} are forward voltages (≈ 0.7 V and ≈ 0.5 V respectively), their difference is small (≈ 0.2 V)
- The transistor behaves as a closed switch: small voltage drop, large current determined by external circuit

14.7 BJT Output Characteristics

The family of curves shows I_C as a function of V_{CE} for different base currents I_B.



Three distinct regions are visible:

- **Cutoff region** ($V_{BE} < 0.6$ **V**): Both junctions reverse-biased. $I_C \approx 0$, transistor is OFF. This is the **open switch** state.
- **Active region** ($V_{CE} > V_{CE,sat}$): EB forward, BC reverse. Here $I_C = \beta I_B$, independent of V_{CE} (ideal current source). This is the **amplification region**.
- **Saturation region** ($V_{CE} < V_{CE,sat}$): Both junctions forward-biased. I_C is no longer controlled by I_B; the transistor behaves like a **closed switch** (small V_{CE}, large I_C).

14.8 The Load Line: Finding the Operating Point

When a BJT is used in an amplifier circuit, it is biased with resistors and power supplies. The **load line** represents all possible combinations of I_C and V_{CE} allowed by the external circuit:

$$I_C = \frac{V_{CC} - V_{CE}}{R_C}$$

Choosing the Q-Point for Amplification

The intersection of the load line with the transistor's output characteristic for a given I_B is the **Q-point** (quiescent point). For linear amplification:

- The Q-point should be centered in the active region
- Far enough from cutoff to avoid distortion of negative signal swings
- Far enough from saturation to avoid distortion of positive signal swings
- The signal should not drive the transistor into cutoff or saturation

When a small AC signal is superimposed on the DC bias:

- The base current varies around its DC value ($I_B \pm i_b$)
- The collector current varies around its DC value ($I_C \pm i_c$)
- The voltage v_{CE} varies according to the load line
- The ratio $i_c/i_b = \beta$ gives the current gain

14.9 Summary: The Three-Terminal Amplifier

Key BJT Equations for Active Mode

Current relationships:

$$\boxed{I_E = I_B + I_C}$$

$$\boxed{I_C = \beta I_B}$$

$$\boxed{I_C = \alpha I_E}$$

Current gain relationships:

$$\boxed{\beta = \frac{\alpha}{1-\alpha}}$$

$$\boxed{\alpha = \frac{\beta}{\beta+1}}$$

Shockley equation for active region:

$$\boxed{I_C = I_S\left(e^{V_{BE}/V_T} - 1\right)}$$

Emitter-base diode equation:

$$\boxed{I_B = \frac{I_S}{\beta}\left(e^{V_{BE}/V_T} - 1\right)}$$

The BJT solves the two-terminal amplification problem by providing a third terminal (the base) that controls the current between emitter and collector. The base current is small because it only needs to replenish the small fraction of carriers lost to recombination. This small control current (I_B) regulates a much larger output current (I_C), giving β typically between 50 and 500. The transistor's three distinct regions—cutoff (OFF), active (amplification), and saturation (ON)—enable both linear amplifiers and digital switches, making the BJT one of the most versatile devices in electronics.